

JAVA LONGEST SUBSTRING PROGRAM

Object-Oriented Programming Practical Record – Program 6

1. Aim

To write a Java program to find the longest substring of a given string that contains no repeated characters.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named **Program6.java**.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Enter a string when prompted.
6. The program checks substrings starting from each character.
7. Repeated characters terminate the current substring.
8. Display the longest substring and its length.

3. Algorithm

1. Start the program.
2. Create a Scanner object to read a string from the user.
3. Initialize the variable **longest** as an empty string.
4. Use an outer loop to select every possible starting position.
5. For each starting position, initialize **current** as an empty string.
6. Use an inner loop to add characters while they are not repeated.
7. If the current character already exists in **current**, stop that substring.
8. Update **longest** whenever the current substring is longer.
9. Display the longest substring and its length.
10. Close the Scanner and stop the program.

4. Explanation of the Program

Scanner: Reads the string entered by the user.

Outer loop: Selects each character as the starting point of a possible substring.

Inner loop: Builds the substring character by character until a repeated character is found.

indexOf(ch): Checks whether the current character is already present in the current substring.

longest: Stores the longest substring found during the search.

Output: Prints the longest substring and its number of characters.

5. Logic / Flow

Step	Operation
1	Read input string
2	Choose a starting character
3	Build current substring
4	Check for repeated character
5	Update longest substring if needed
6	Repeat for all starting positions
7	Print result

6. Java Source Code

```
import java.util.Scanner;

public class Program6 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String longest = "";

        for (int i = 0; i < str.length(); i++) {

            String current = "";

            for (int j = i; j < str.length(); j++) {

                char ch = str.charAt(j);

                if (current.indexOf(ch) != -1) {
                    break;
                }

                current = current + ch;

                if (current.length() > longest.length()) {
                    longest = current;
                }
            }
        }

        System.out.println("Longest substring = " + longest);
        System.out.println("Length = " + longest.length());

        sc.close();
    }
}
```


6. Java Source Code (continued)

```
import java.util.Scanner;

public class Program6 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String longest = "";

        for (int i = 0; i < str.length(); i++) {

            String current = "";

            for (int j = i; j < str.length(); j++) {

                char ch = str.charAt(j);

                if (current.indexOf(ch) != -1) {
                    break;
                }

                current = current + ch;

                if (current.length() > longest.length()) {
                    longest = current;
                }
            }
        }

        System.out.println("Longest substring = " + longest);
        System.out.println("Length = " + longest.length());

        sc.close();
    }
}
```

7. Sample Output

```
Enter a string: abcabcbb
Longest substring = abc
Length = 3
```

8. Result

Thus, the Java program successfully finds and displays the longest substring without repeated characters along with its length.

Example Explanation

For the input **abcabcbb**, the longest substring without repeating characters is **abc**, and its length is **3**.

Student Signature

Faculty Signature
